Project Assignment: A microcontroller-based Voltage Protector Device that protects electrical appliances from sudden power-grid voltage surges or overcurrent faults, with an auto-disconnect feature and pc monitoring and control support.

Design Approach: I used an ESP32 as the brain of the project. Since using main's was forbidden so I used 12 volts DC to simulate AC. I have used a boost converter to change the dc voltage level. The esp can sense voltage upto 3.3 volt, so I have used a voltage divider by using 200k and 10k resistors to feed the esp. Basically, I divide the 20 V grid down by 21 with the 200k/10k divider, so the ESP32 sees around 0.95 V. Then I multiply back: pin millivolts x 21 gives the bench voltage, and x 11.5 more (because 20 V means 230 V) gives the grid value. So 952 mV x 21 x 11.5 / 1000 comes to 230 volts. For measuring the current, I have used 10 ohm 3watt resistor as a shunt.. 0.32 A makes 3.2 V across it, and the 20k/10k divider brings that down by 3.2 to about 1 volt at the pin. So pin millivolts x 3.2 / 10 / 1000. That means 1000 mV gives 0.32 amp.

Implementation Process: For controlling the load connection, I have used a single-channel relay module. After sudden power-grid voltage surges, the module disconnects the relay for 10 seconds before reconnecting. Also, I have soldered a 10k pot to the boost converter for ease of control. For observing the voltage, current value, and behaviour I have used a 12*2 LCD module with a PCF8575 I2C converter. I have also used a buzzer for notification and two led. Green indicates normal operation, and red indicates faults. For communication with PC I have used HW 519 rs485 adapter. RS485 is best for industrial usages for its differential pair communication, which provides less interference.

Results: The relay disconnects the loads as soon as a power surge happens. For the threshold, I set 250 volts as the upper limit and 180 volts as the lower limit. Also I have set 400ma as the current limit. To mimic high current consumption, I have used a mini n20 dc motor from my old line follower. When I try to hold the wheel the current consumption of the motor increases, and the microcontroller detects it and disconnect the load from the power source using relay. Also I can change the threshold values of voltage and current from the pc using rs485 adapter from pc serial monitor. For that I have used the software Serial Debug Assistant with baud rate of 115200.

Areas of Improvement: The inrush current from the motor can trip the overcurrent detection module, which can be solved by adding a calculated delay for the load in the code. Also the project can be scaled for real world application using potential transformer and current transformer to calculate real AC load with magnetic conductor instead of relay for switching.\

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10-10-2026